Three Variables Suffice for Commuting Row Contractions on $\mathbb{C}^4$

Automated mathematics run `fa_banach_001`, lane 12

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Status. Substantial partial result, likely valid. We prove that the optimal row-contractive von Neumann constants satisfy $C_{d,4} = C_{3,4}$ for every $d \geq 3$. Thus the source question in matrix dimension four reduces completely to deciding whether $C_{2,4} < C_{3,4}$. That final comparison remains open.

1 Source question and result

Hartz, Richter, and Shalit prove a dimension-free-in-d von Neumann inequality for commuting row contractions of fixed matrix size [1]. For $d, n \geq 1$, let $C_{d,n}$ be the least constant such that

$$\|p(T)\| \leq C_{d,n} \sup_{z \in \mathbb{B}_d} |p(z)| \tag{1}$$

for every polynomial $p \in \mathbb{C}[z_1, \ldots, z_d]$ and every commuting tuple $T = (T_1, \ldots, T_d)$ on $\mathbb{C}^n$ satisfying

$$\sum_{j=1}^{d} T_j T_j^* \leq I. \tag{2}$$

In Remark 4.8 on page 12 they state that they cannot determine whether

$$C_{2,n} < C_{d,n} \tag{3}$$

for any $n > 2$.

Piekarz subsequently settled the first possible matrix size, proving $C_{d,3} = C_{2,3}$ for all $d \geq 2$ [2]. The following theorem handles all variable dimensions beyond two in the next matrix size.

Theorem 1 (Dimension-four stabilization)**.** *For every $d \geq 3$,*

$$\boxed{C_{d,4} = C_{3,4}.} \tag{4}$$

Consequently, $\sup_d C_{d,4} = C_{3,4}$, and a strict dependence on d in matrix dimension four can only be the single jump $C_{2,4} < C_{3,4}$.

2 Idea of the proof

Move one joint eigenvalue of a strict tuple to the origin with a ball automorphism. If there is more than one joint eigenvalue, primary decomposition and Schur's sharp dimension bound for commutative matrix algebras show that the transformed coordinate matrices span at most three dimensions. A unitary rotation of the variables then kills all later coordinates.

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Remark 4.8. It follows from the incomparability of the norms $\|\cdot\|_{\mathrm{Mult}(H^2_d)}$ and $\|\cdot\|_\infty$ (see the discussion in the introduction) that $C_{d,n} \xrightarrow{n\to\infty} \infty$ for all $d \geq 2$. In fact, considering the compression of the tuple M_z on H^2_d to the space of all polynomials of degree at most k and using computations done in the proof of [4, Theorem 3.3], one can show that for $d \geq 2$ and $k \geq 1$,

$$C_{d,n} \geq (2\pi)^{\frac{d-1}{4}} d^{-\frac{1}{4}} k^{\frac{d-1}{4}},$$

where $n = \binom{dk+d}{d}$ is the dimension of the space of polynomials of degree at most k in d variables. In particular, for sufficiently large n, we have

$$C_{d,n} \geq C_{2,n} \geq n^{\frac{1}{8}}.$$

Unfortunately, the gap between these lower bounds and the upper bounds given by Theorem 4.7 is huge, and we are not able to determine whether there is a strict inequality $C_{2,n} < C_{d,n}$ for any $n > 2$.

Figure 1: Remark 4.8 on page 12 of arXiv:2109.08550.

If there is only one joint eigenvalue, all transformed coordinates are commuting nilpotent 4×4 matrices. Their span again has dimension at most four. The only apparent obstruction is the extremal four-dimensional case. Maximality then forces this span to be an algebra. An elementary centralizer argument in the strictly upper triangular algebra proves that every product in this algebra is zero.

That exceptional square-zero case is harmless analytically. The functional calculus retains only a bounded holomorphic function's value and gradient at the origin. Schwarz–Pick bounds the gradient, and a one-variable disk automorphism matches the same first jet. The classical von Neumann inequality therefore gives constant one. In every case, either three variables suffice or the tuple satisfies a stronger constant-one estimate.

3 The maximal nilpotent algebra

We use Schur's theorem in the form used in Lemma 4.5 of [1]: a commutative subalgebra of $M_r(\mathbb{C})$ has dimension at most $\lfloor r^2/4\rfloor + 1$.

Lemma 2 (Extremal nilpotent lemma)**.** *Let $L \subset M_4(\mathbb{C})$ be a four-dimensional linear space of pairwise commuting nilpotent matrices. Then*

$$L^2 = \{0\}.$$

Proof. Let $\mathcal{A} = \mathrm{Alg}(I, L)$ be the unital algebra generated by L. Schur's bound gives $\dim\mathcal{A} \leq 5$, while $\mathbb{C}I + L$ already has dimension five. Thus $\mathcal{A} = \mathbb{C}I + L$. If $x, y \in L$, then xy is nilpotent because x and y commute. Write $xy = \alpha I + z$ with $z \in L$. If $\alpha \neq 0$, then $\alpha I + z$ is invertible, a contradiction. Hence $xy \in L$, so L is an algebra.

A commuting family of complex nilpotents is simultaneously strictly upper triangularizable. In a common basis write

$$x = \begin{pmatrix} 0 & a & b & c \\ 0 & 0 & d & e \\ 0 & 0 & 0 & f \\ 0 & 0 & 0 & 0 \end{pmatrix} \in L. \tag{5}$$

The matrix unit E_{14} annihilates every strictly upper triangular 4×4 matrix on both sides. If $E_{14} \notin L$, then $\mathbb{C}I + L + \mathbb{C}E_{14}$ is a six-dimensional commutative algebra, contrary to Schur's bound. Therefore $E_{14} \in L$.

Suppose that some $x \in L$ has $x^2 \neq 0$. For

$$y = \begin{pmatrix} 0 & A & B & C \\ 0 & 0 & D & E \\ 0 & 0 & 0 & F \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

the equation $xy = yx$ is equivalent to

$$aD = Ad, \qquad dF = Df, \qquad aE + bF = Ae + Bf. \tag{6}$$

First assume $d \neq 0$. The first two equations determine A and F from D. Substitution into the third gives one nonzero linear relation among B, D, E. Indeed, if this relation vanished identically then $a = f = 0$, which would imply $x^2 = 0$. With the free central coordinate C, the centralizer of x inside the strictly upper triangular matrices therefore has dimension at most three. It cannot contain the four-dimensional space L.

It remains that $d = 0$. Now $x^2 = (ae + bf)E_{14} \neq 0$, so $(a, f) \neq (0, 0)$, and the first two equations in (6) force $D = 0$. On $\mathbb{C}^4$ define the nondegenerate alternating form

$$\Omega((a, b, e, f), (A, B, E, F)) = aE + bF - Ae - Bf.$$

Project the centralizer coordinates to (A, B, E, F). Its kernel on L is exactly $\mathbb{C}E_{14}$, so the image of L has dimension three. Pairwise commutativity says that this image is Ω-isotropic. But an isotropic subspace of a four-dimensional symplectic space has dimension at most two, a contradiction.

We have proved $x^2 = 0$ for every $x \in L$. Since L is linear and commutative, $0 = (x + y)^2 = x^2 + 2xy + y^2$ gives $xy = 0$ for all $x, y \in L$. $\square$

4 The spectral and variable reduction

Lemma 3 (Four-dimensional dichotomy)**.** *Let $T = (T_1, \ldots, T_d)$ be a strict commuting row contraction on $\mathbb{C}^4$. There are a ball automorphism $\theta \in \mathrm{Aut}(\mathbb{B}_d)$ and a unitary rotation of the variables such that $S = \theta(T)$ is a strict commuting row contraction and either*

(i) *$S_j = 0$ for every $j > 3$; or*

(ii) *$S_i S_j = 0$ for every $1 \leq i, j \leq d$.*

Proof. Choose $\lambda \in \sigma(T)$ and let θ take λ to zero. Ball automorphisms preserve strict commuting row contractions. For completeness, for the standard involution φ_a this follows from the defect identity

$$\begin{aligned} I - \varphi_a(T)\varphi_a(T)^* = (1 - \|a\|^2) \left(I - \sum_j \overline{a_j} T_j \right)^{-1} \left(I - \sum_j T_j T_j^* \right) \\ \cdot \left(I - \sum_j a_j T_j^* \right)^{-1} > 0. \end{aligned} \tag{7}$$

Unitary rotations preserve the same property. The inverse automorphism recovers T from S, so the generated algebras, and hence the primary block sizes, are preserved.

Let the joint generalized eigenspace blocks of S have dimensions $r_1, \dots, r_m$, with $\sum r_k = 4$. On block k write

$$S_i|_k = \lambda_i^{(k)} I + N_i^{(k)},$$

where the $N_i^{(k)}$ are commuting nilpotents. Schur's bound applied to their generated unital algebra gives

$$\dim \operatorname{span}_i\{N_i^{(k)}\} \leq \left\lfloor \frac{r_k^2}{4} \right\rfloor. \tag{8}$$

One spectral point is zero, so the scalar block patterns span at most $m-1$ dimensions. Consequently,

$$\dim \operatorname{span}\{S_1, \dots, S_d\} \leq (m-1) + \sum_{k=1}^{m} \left\lfloor \frac{r_k^2}{4} \right\rfloor. \tag{9}$$

If $m \geq 2$, the four possible partitions give respectively

$$3+1:3, \qquad 2+2:3, \qquad 2+1+1:3, \qquad 1+1+1+1:3.$$

Thus the coordinate span has dimension at most three.

If $m = 1$, every S_i is nilpotent and Schur's bound gives $\dim \operatorname{span}\{S_i\} \leq 4$. Dimension at most three gives (i). In dimension four, Lemma 2 gives (ii). Finally, whenever the coordinate span has dimension at most three, choose an orthonormal basis of the coefficient space adapted to the kernel of $c \mapsto \sum c_i S_i$. The associated unitary variable rotation kills all coordinates after the third. $\square$

5 The square-zero case

Lemma 4 (First-jet von Neumann inequality)**.** *Let $S = (S_1, \dots, S_d)$ be a commuting row contraction with $S_i S_j = 0$ for all i, j. If $q : \mathbb{B}_d \to \mathbb{D}$ is holomorphic, then*

$$\|q(S)\| \leq 1.$$

Proof. Put $a = q(0)$, $b = (\partial_1 q(0), \dots, \partial_d q(0))$, and $r = \|b\|_2$. The infinitesimal Schwarz–Pick lemma on the ball [3] gives

$$r \leq 1 - |a|^2. \tag{10}$$

All terms of degree at least two vanish in the functional calculus, hence

$$q(S) = aI + \sum_j b_j S_j. \tag{11}$$

If $r = 0$ there is nothing to prove. Otherwise let

$$A = \sum_j \frac{b_j}{r} S_j, \qquad s = \frac{r}{1 - |a|^2} \leq 1.$$

Row contractivity gives $\|A\| \leq 1$, and $A^2 = 0$. The disk Schur function

$$h(z) = \frac{a + sz}{1 + \overline{a} s z}$$

satisfies $h(0) = a$ and $h'(0) = r$. Therefore $h(A) = aI + rA = q(S)$, and the classical von Neumann inequality (see, e.g., [1, Section 1]) gives $\|q(S)\| \leq \|h\|_{\infty,\mathbb{D}} \leq 1$. If $|a| = 1$, the maximum principle makes q constant, which is the omitted degenerate case. $\square$

6 Proof of the stabilization theorem

Proof of Theorem 1. Fix $d \geq 3$, a commuting row contraction T on $\mathbb{C}^4$, and a polynomial p. Applying the argument first to ρT for $0 < \rho < 1$ and then letting $\rho \uparrow 1$, it suffices to treat strict tuples.

Apply Lemma 3, including its unitary rotation, and put $S = \theta(T)$ and $q = p \circ \theta^{-1}$. The superposition property of the holomorphic functional calculus gives

$$p(T) = q(S), \qquad \|q\|_{\infty,\mathbb{B}_d} = \|p\|_{\infty,\mathbb{B}_d}. \tag{12}$$

The rational function q is holomorphic on a neighborhood of $\overline{\mathbb{B}_d}$.

In case (i), let $\iota : \mathbb{C}^3 \to \mathbb{C}^d$ be the coordinate inclusion. Since $S_j = 0$ for $j > 3$, functional calculus gives $q(S) = (q \circ \iota)(S_1, S_2, S_3)$. Uniform polynomial approximation of $q \circ \iota$ on $\overline{\mathbb{B}_3}$ and the definition of $C_{3,4}$ yield

$$\begin{aligned} \|p(T)\| = \|q(S)\| &\leq C_{3,4}\|q \circ \iota\|_{\infty,\mathbb{B}_3} \\ &\leq C_{3,4}\|p\|_{\infty,\mathbb{B}_d}. \end{aligned}$$

In case (ii), normalize q by its supremum norm and apply Lemma 4; this gives the stronger bound $\|p(T)\| \leq \|p\|_{\infty,\mathbb{B}_d}$. Hence $C_{d,4} \leq C_{3,4}$.

Conversely, pad every three-tuple with zero coordinates and regard every three-variable polynomial as a d-variable polynomial. Its supremum on $\mathbb{B}_d$ equals its supremum on $\mathbb{B}_3$, so $C_{d,4} \geq C_{3,4}$. This proves (4). Since $C_{1,4} = 1 \leq C_{2,4} \leq C_{3,4}$, the assertion about the supremum follows as well. □

7 Verification, novelty boundary, and review focus

The exact checker in `code/verify_dimension_four_nilpotent_algebra.py` enumerates all 11,011 four-dimensional subspaces of the six-dimensional strictly upper triangular algebra over $\mathbb{F}_3$. It finds one commutative subalgebra and verifies that it is square-zero, ending with `ALL FINITE-FIELD NILPOTENT-ALGEBRA CHECKS PASSED`. This is an exact finite stress test, not a proof over $\mathbb{C}$.

The literature check was bounded. On August 9, 2026, exact and close arXiv searches used $C_{d,4}$, $c_{d,4}$, "4×4 row contractive von Neumann," and close variants. The source and Piekarz papers were inspected; no statement of Theorem 1 was found. This is a bounded novelty signal, not an exhaustive certification.

Human review should prioritize the maximality-to-algebra step, both centralizer cases, the block dimension count, the defect identity, and the first-jet matching. Recommended status after review: `substantial_partial_result_likely_valid`.

8 Limitation

The theorem completes the variable-stabilization program for matrix size four but does not answer the source question there: it remains possible that

$$C_{2,4} < C_{3,4}.$$

No numerical value or sharp bound for either constant is claimed.

References

[1] M. Hartz, S. Richter, and O. M. Shalit, *von Neumann's inequality for row contractive matrix tuples*, Math. Z. **301** (2022), 3877–3894; arXiv:2109.08550.

[2] D. Piekarz, *The von Neumann inequality for* 3×3 *matrices in the unit Euclidean ball*, Israel J. Math. (2025); arXiv:2310.12908.

[3] W. Rudin, *Function Theory in the Unit Ball of* $\mathbb{C}^n$, Springer-Verlag, New York, 1980.